R-107, First Floor, Metro Tower, Vijay Nagar, Indore (M.P.) 452010

Cypher Analytica — SQL and Database Management Course

Level: Beginner to Intermediate, with an introduction to advanced SQL
Proposed duration: 60 instructor-led hours + 30 hours of independent practice
Suggested schedule: 10 weeks × 3 sessions × 2 hours
Suitable for: Students, aspiring data analysts, business analysts, developers, and working professionals
Prerequisites: Basic computer knowledge; no programming experience required.

Primary platform: MySQL
Teaching approach: Concept explanation, live queries, business exercises, assignments, and a database project. Differences from PostgreSQL and SQL Server can be highlighted where relevant.

Course objectives

By the end of the course, learners should be able to:

- Design relational tables with appropriate keys and constraints.
- Retrieve, filter, combine, and summarize business data.
- Write subqueries, CTEs, and window-function queries.
- Create views, stored procedures, functions, and triggers.
- Apply transactions and database permissions.
- Identify common query-performance issues.
- Complete and explain a practical SQL project.

Module-wise curriculum

Module	Topics covered	Hours
1. Database Fundamentals and Setup	Databases and DBMS; relational concepts; tables, rows, and columns; primary and foreign keys; client–server basics; MySQL and Workbench setup; SQL statement categories	3
2. Database Design and DDL	Entities and relationships; data types; CREATE, ALTER, DROP, and TRUNCATE; PRIMARY KEY, FOREIGN KEY,	5

Module	Topics covered	Hours
	UNIQUE, NOT NULL, CHECK, and DEFAULT; introductory normalization through 3NF	
3. Data Manipulation and Basic Queries	INSERT, UPDATE, DELETE; SELECT; aliases; DISTINCT; calculated columns; safe update and delete practices; importing and exporting CSV data	4
4. Filtering, Sorting, and NULL Handling	WHERE; comparison and logical operators; BETWEEN, IN, and LIKE; IS NULL; ORDER BY; LIMIT; operator precedence; three-valued logic and common NULL mistakes	4
5. Built-in Functions and Conditional Logic	String, numeric, and date functions; type conversion; CASE; COALESCE and NULLIF; text cleaning; date calculations; handling missing values	5
6. Aggregation and Business Summaries	COUNT, SUM, AVG, MIN, and MAX; GROUP BY; HAVING; conditional aggregation; distinct counts; aggregation grain; treatment of NULLs	4
7. Joins and Set Operations	Inner, left, right, cross, and self joins; multi-table joins; unmatched records; join cardinality; avoiding double counting; UNION versus UNION ALL; full outer join emulation in MySQL	6
8. Subqueries and Common Table Expressions	Scalar and multi-row subqueries; correlated subqueries; EXISTS and NOT EXISTS; derived tables; nonrecursive CTEs; introductory recursive CTEs; breaking complex queries into steps	5
9. Window Functions and Analytical SQL	OVER, PARTITION BY, and window ordering; ROW_NUMBER, RANK, and DENSE_RANK; LAG and LEAD; aggregate windows; window frames; running totals; moving averages; top-N per group	6
10. Views and Database Programming	Views; stored procedures; parameters; user-defined stored functions; variables and conditional logic; introductory triggers; choosing the appropriate database object	6

Module	Topics covered	Hours
11. Transactions, Security, and Performance	ACID concepts; START TRANSACTION, COMMIT, ROLLBACK, and savepoints; autocommit and implicit-commit awareness; GRANT and REVOKE; least privilege; parameterized-query concepts; indexes; reading basic EXPLAIN output	6
12. Capstone Project and Assessment	Business requirements; schema design; data loading; analytical queries; result validation; query improvement; documentation; presentation and viva	6
Total	**Guided instruction, practical work, and assessment**	**60**

Practical exercises

Learning area	Suggested task
Database design	Design student, course, batch, and enrollment tables
Data manipulation	Insert records and safely update or remove selected entries
Filtering and functions	Find employees by department, joining date, and salary range
Aggregation	Calculate monthly revenue, order counts, and average order value
Joins	Combine customers, orders, and products without inflating totals
Subqueries and CTEs	Identify employees earning above their department average
Window functions	Rank products, calculate running sales, and compare monthly performance
Data quality	Detect duplicates, missing values, and unmatched reference records
Database programming	Create a parameterized report procedure and a simple audit trigger
Transactions	Perform related updates and verify rollback behavior

Learning area	Suggested task
Performance	Compare a query's execution plan before and after adding an appropriate index

Capstone project options

1. **Retail Sales Analysis:** Customers, products, orders, and order items; monthly revenue, customer contribution, product ranking, and repeat purchases.
2. **Institute Management Database:** Students, courses, batches, enrollments, attendance, and payments; enrollment trends, attendance summaries, and outstanding fees.
3. **Employee and Department Analysis:** Employees, departments, roles, and salary records; headcount, tenure, department comparisons, and salary ranking.
4. **Inventory Management Database:** Products, suppliers, purchases, and stock movements; stock balances, low-stock items, and supplier summaries.

Mandatory project deliverables

- An entity–relationship diagram and data dictionary.
- SQL scripts to create tables, constraints, and sample data.
- At least 15 business queries using joins, aggregation, CTEs, and window functions.
- A view and a stored procedure serving clear business requirements.
- Validation checks for duplicates, NULLs, relationships, and report totals.
- One documented query-performance improvement.
- A README explaining execution steps, assumptions, and findings.

Assessment Structure

Component	Weight
Weekly SQL assignments	25%
Practical query assessments	25%
Database design exercise	15%

Component	Weight
Capstone project	25%
Technical viva and query explanation	10%
Total	**100%**

Certification Criteria: Minimum 60% overall, submission of mandatory assignments, and successful demonstration of the capstone project.

Interview preparation

Practice explaining WHERE versus HAVING, joins versus subqueries, UNION versus UNION ALL, ranking functions, keys and constraints, NULL behavior, duplicate detection, and transaction handling. Include timed business-query exercises and debugging incorrect results.

Progression: This course provides a SQL foundation for data analytics, BI, backend development, and data engineering. Database administration, advanced performance tuning, and production operations require further training.